

- ▶ **Upgrading gets easier:** Upgrading software on servers is a tough task. Sometimes you just need to shut down the machine and when it's about servers which make a business run, it's not always possible. With virtualization, you can simply clone the machine and upgrade it and then replace it with the current one. Well, it's not so easy as it seems but comparatively it's a lot easier. Again, if you need to restart a machine, all you have to do is power off the virtual machine and bring it back on! So much easier compared to restarting the real machines (with virtualization, the restarting job can be done remotely).
- ▶ **Scalability:** It simply means the ability to expand. So if today you're happy with one server, tomorrow the business will grow and you'll need two servers, and then three and then probably even more! Virtualization makes scaling a whole lot easier. Let's consider a scenario: you have one server which runs seven virtual machines each doing its own job and one of them is a machine that handles email. Now suppose lots of email starts coming in and you're in need of sending a bunch of mail as well (imagine Facebook – it sends a huge number of emails per second for all those likes, shares and comments). That's when you're in need of another email server. You can easily create a clone of the first email server virtual machine and use it to create another one on a different server and configure the setup! Now if you want to move the email server virtual machine to another physical server, you can easily do that; after all, you just need to copy a virtual machine!
- ▶ **Saving software costs:** If you have to install more servers which will run more than one virtual machine, virtualization will help save money. How? Let's suppose, you have five physical servers with Windows Server as the host OS on them; they run multiple VMs on each of them and are fully loaded. If you need to create more virtual machines, you'll need a sixth physical server. Now, if you were to install Windows Server on the sixth machine as well, it would cost you quite a lot. But you don't need to spend! The guest OS (the one running inside the virtualization software) never knows that it's not running on the real hardware. So you can just install Linux on the sixth server and then create the virtual machines inside it and it will work just as well!
- ▶ **Resource sharing:** As mentioned before, if you wanted to create a new email server with a virtual machine, you could just create a copy of the first virtual machine running the email server and it would work fine. However, both email servers need to share the database where the emails